Chris Pecunies

Software Engineer | Backend and Distributed Systems

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Software engineer building cloud applications, distributed services, and developer tooling. Experience includes AWS application development, Go and Rust backend systems, infrastructure automation, and blockchain operations. Independent systems work spans compiler and language development and a WebAssembly runtime in Zig.

Technical Skills

Languages: Rust, Go, Python, TypeScript, Zig, C/C++, SQL

Backend and systems: Distributed systems, WebAssembly, gRPC, Protocol Buffers, FastAPI, PostgreSQL, Linux

Cloud and delivery: AWS, GCP, Azure, AWS CDK, Terraform, Ansible, Docker, Kubernetes, CI/CD, Argo CD, Prometheus, Grafana

Experience

Amazon Leo Current

Software Development Engineer II | Seattle, WA

- Develop applications and contribute to build, test, and CI/CD workflows for Amazon Leo.

HashGraph Sep 2025 - Nov 2025

DevOps Engineer | Remote / Seattle, WA

- Managed blockchain infrastructure across 8+ environments, using Argo CD and Grafana to support deployments and operational visibility.
- Built artifact verification and cryptographic integrity checks into deployment workflows.
- Provisioned GCP infrastructure with Terraform and Ansible; tuned PromQL alerts and dashboards to reduce false positives.

WiseBlocks LLC Jun 2022 - Apr 2024

Software Engineer | Hybrid / Golden, CO

- Developed a distributed transaction database in Go and integrated a Rust WebAssembly virtual machine for smart contract execution.
- Designed serialized data structures and gRPC/Protocol Buffers APIs for data exchange between network nodes.
- Automated cloud infrastructure and delivery workflows with Ansible and Terraform, with Prometheus and Grafana for monitoring.
- Built Next.js interfaces for node and block visualization and a FastAPI service for delivering data to nodes.

Experience Continued

Impresys Software Corporation Sep 2019 - May 2022

AWS Consultant | Seattle, WA

- Developed and delivered AWS and Azure technical training on cloud architecture, infrastructure automation, and CI/CD.
- Collaborated with AWS engineers on AWS CDK demonstrations and automated environment provisioning.
- Modernized legacy production workflows with Python, Qt/QML, and OpenCV.

University of Washington Jun 2018 - Apr 2021

Research Assistant | Seattle, WA

- Built containerized web applications and REST services using AWS, React, Django, FastAPI, PostgreSQL, and Docker.
- Developed Python tools for scientific simulation, data modeling, and computer vision; mentored undergraduate researchers in machine learning and data analysis.

Selected Projects

Idol Ongoing

Independent compiler and language project

- Developing a programming language and compiler, with work on graph-based representations, program analysis, and code generation.

wart May 2025 - Present

WebAssembly runtime in Zig

- Developing a WebAssembly runtime in Zig, with work on instruction dispatch, memory layout, runtime performance, and WASI support.

Marketplace Aggregator on AWS Apr 2026 - Present

moe.pecunies.com

- Built a serverless marketplace aggregator with AWS Lambda, Step Functions, DynamoDB, SQS, API Gateway, CloudFront, and AWS CDK.
- Designed retry, state-management, and idempotency mechanisms to handle external API failures, rate limits, and duplicate listing prevention.

Education

University of Washington Jun 2019

B.S. Materials Science and Engineering
Nanotechnology and Molecular Engineering concentration